Empirical parameter values for a Bd compartmental ODE model

Seven parameters governing chytrid fungus dynamics in amphibians can be estimated from published laboratory experiments, field studies, and existing mathematical models, though values span wide ranges depending on host species, Bd strain, and environmental context. The most extensively parameterized Bd model remains Briggs et al. (2010), which uses a zoospore-pool framework rather than a classical SIR structure. Translating those within-host load dynamics into your compartmental structure requires careful interpretation. Below is a comprehensive compilation of every empirical estimate found, organized by parameter, with exact sources and context.

α — Transmission from infected to susceptible frogs spans orders of magnitude

Direct estimation of a simple density-dependent transmission rate α (as in $dI = \alpha \cdot I \cdot S$) for Bd is complicated by the fact that transmission occurs via a free-living zoospore stage rather than direct contact. The most rigorous approach comes from the Briggs et al. (2010) zoospore pool model, which decomposes transmission into zoospore encounter rate, infection probability, and zoospore production.

Briggs et al. (2010), PNAS 107:9695–9700, DOI: 10.1073/pnas.0912886107 — Species: *Rana muscosa*/*R. sierrae*, Sierra Nevada, California. Model equations: $dS_i/dt = (\beta/V)Z + \eta\nu f S_i - \sigma S_i$, where transmission is mediated by an environmental zoospore pool Z. (nih) Key parameter values from Figures 2–4:

- Zoospore encounter rate: **γ = 0.01** (unit volume·day^{-1}), also explored at 10^{-6}, 10^{-4}, 10^{-3}
- Infection success fraction: **ν = 0.05–0.1** (dimensionless)
- Composite transmission rate: **β = νγ** (ranging ~5×10^{-4} to 10^{-3} at default γ)
- Self-reinfection fraction: **f = 0.05–0.15**

Quote: "Zoospores from the pool encounter frogs at rate γ, and a fraction, ν, of those zoospores successfully infect the frog skin and become sporangia, such that the rate of successful transmission is β = νγ." (nih) (PubMed Central)

Rachowicz & Briggs (2007), J. Anim. Ecol. 76:711–721, DOI: 10.1111/j.1365-2656.2007.01256.x — Species: *R. muscosa*. Used maximum likelihood to fit transmission functions. (Wiley) Found **density-dependent transmission** best fit experimental data: proportion of tadpoles becoming infected increased with number of previously infected tadpoles present. This paper validates the α·I·S form for Bd transmission.

Wilber et al. (2016), Methods Ecol. Evol. 7:1182–1194, DOI: 10.1111/2041-210X.12561 — Species: *R. muscosa*. Integral Projection Model framework. PubMed Temperature-dependent probability of acquiring infection from environment: **ω = 0.22–0.6** per time step. Density-dependent transmission parameter explored across **β = 0 to 1.17×10^{-3}**. PubMed Central

Wilber et al. (2017), Ecol. Lett. 20:1169–1181, DOI: 10.1111/ele.12814 — Mesocosm transmission experiment with *R. muscosa*. Density treatments of 1, 4, 8, 16 uninfected adults per ~1 m^3 mesocosm, with 5 infected tadpoles. Monitored over 74 days at 3-day time steps. Found **density-dependent transmission + environmental zoospore pool** provided best fit. Including the environmental pool significantly increased R_0. University of Colorado Bou… University of Colorado Bou…

Mitchell et al. (2008), Proc. R. Soc. B 275:329–334, DOI: 10.1098/rspb.2007.1356 — Species: *Bufo bufo*. SI model with free-living zoospore stage. Transmission parameter **$\nu \approx 1.8–2.4 \times 10^{-8}$**. Zoospore release rate $\rho = 10^5$ to 6×10^6 zoospores yr^{-1} per infected individual. PubMed

For your model's α (simplified direct transmission): To collapse the zoospore-pool mechanism into $\alpha \cdot I \cdot S$, a rough estimate can be derived. Briggs' model at default parameters (γ=0.01, ν=0.1, η=17.5 zoospores/day per sporangium, μ=1 day^{-1} zoospore mortality, PubMed Central K=100 frogs) gives transmission dynamics where populations crash within 2–3 years. A plausible range for a simplified **$\alpha \approx 10^{-5}$ to 10^{-3} per frog per day**, depending on host density, Bd strain virulence, and environmental context. DiRenzo et al. (2014, PLoS ONE 9:e93356) measured a **35.4% daily increase** in infection intensity on *Atelopus zeteki* (a supershedder species), providing a within-host growth rate benchmark. ResearchGate Mendeley

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β — Infected frogs succumb at rates of ~0.01–0.05 per day, species-dependent

Disease-induced mortality rate is highly species-specific and load-dependent. The **10,000 zoospore equivalents (ZE) threshold** identified by Vredenburg et al. (2010) is the most widely cited benchmark for population-level collapse. University of Colorado Bou…

Vredenburg et al. (2010), PNAS 107:9689–9694, DOI: 10.1073/pnas.0914111107 — Species: *R. muscosa*/*R. sierrae*. "Dynamics of an emerging disease drive large-scale amphibian population extinctions." PNAS PubMed Critical lethal threshold: **mean of 11,775 ± 5,851 ZE/swab** at population collapse. Exceeding ~10,000 ZE consistently resulted in mass mortality. Populations crashed within **2–3 years** of Bd arrival. PubMed PubMed Central Independently validated in *Lithobates areolatus* by Kinney et al. (2011, PLoS ONE 6:e16708).

Murphy et al. (2011), Dis. Aquat. Org. 95:31–42, DOI: 10.3354/dao02336 — Species: *Anaxyrus boreas* (boreal toad). At ~10^6 zoospore dose with continuous water contact: **all toads died within 30 days** → $\beta \approx$ **0.033 day^{-1}**. With dry microenvironments available: mean survival **48 days** at 15°C → $\beta \approx$ **0.021 day^{-1}**. At 18°C with behavioral thermoregulation: survival extended to **115 days**. Previously exposed toads survived **3× longer** (89 vs. 30 days).

DiRenzo et al. (2014), PLoS ONE 9:e93356 — Species: *Atelopus zeteki* (Panamanian golden frog, a supershedder). Time from inoculation to death: **22–33 days (mean 28.88 ± 1.58)** (Mendeley) → β ≈ **0.030–0.045 day⁻¹**. (PLOS) Zoospore shedding rate near death: **~23.55 ZGE per minute**. (ResearchGate) (ResearchGate)

Murray et al. (2009), Conserv. Biol. 23:1242–1252, DOI: 10.1111/j.1523-1739.2009.01211.x — Species: *Litoria pearsoniana* (cascade treefrog, Australia). Endemically infected population. Bd **reduced monthly survival of infected individuals by 38%**. If uninfected monthly survival ≈ 0.90, then infected ≈ 0.56, giving excess mortality ≈ **β ≈ 0.014 day⁻¹**.

Berger et al. (2004), Aust. Vet. J. 82:434–439 — Species: *Mixophyes fasciolatus*. At 17°C and 23°C: **100% mortality**. At 27°C: **50% mortality** (Conservation Evidence) (4/8 survived), survivors cleared infection by 98 days. (ResearchOnline)

Briggs et al. (2010) — In enzootic (persistent) populations of *R. muscosa*, mean adult Bd load was only **220 ± 81 ZE/swab** (median 20.5), and Bd infection status had **no detectable effect on adult survival** (nih) (PubMed) (annual survival 48–87% depending on site). This implies β ≈ 0 for adults in enzootic conditions with low loads.

Recommended range for β: 0.014–0.045 day⁻¹ (corresponding to survival times of ~22–70 days post-infection for susceptible species at moderate-to-high Bd loads). For enzootic, low-load conditions: β may approach 0.

γ — Cadaver and environmental transmission is real but poorly quantified

The R1 compartment (deceased frogs still transmitting Bd) reflects an important biological reality. Bd can persist on dead organic matter and in water, (PNAS) and the related species Bsal has been directly shown to transmit from carcasses.

Bsal carcass data (analog for Bd): Carter et al. (2024, Front. Amphib. Reptile Sci., DOI: 10.3389/famrs.2024.1284608) — For *Batrachochytrium salamandrivorans* on eastern newts: "Following introduction to a tank, carcasses shed approximately **1,850 zoospores in the first 24 hrs, 999 zoospores in the next 24 hrs, and 1,224 spores in the third 24-hr period.**" (Frontiers) These shedding rates resulted in **100% indirect transmission and 83% mortality** of susceptible newts in partitioned cohousing. (Frontiers) By contrast, DiRenzo et al. (2014) estimated live *A. zeteki* shed ~23.55 ZGE/minute (~34,000/day), roughly **500× the carcass shedding rate**. (Frontiers) (ResearchGate)

Johnson & Speare (2003), Emerg. Infect. Dis. 9:922–925 — Bd zoosporangia were observed growing on dead arthropod skeletons and algae in sterile lake water: "Zoosporangia of strain 98-1810/3 visible as transparent spherical bodies growing in lake water on freshwater arthropod and algae." (CDC) This demonstrates Bd saprophytic capacity on non-amphibian organic substrates and, by extension, on dead amphibian tissue.

McMahon et al. (2013), PNAS 110:210–215 — Crayfish maintained Bd infection for **at least 12 weeks** and transmitted Bd to amphibians, demonstrating non-amphibian reservoir transmission. (PubMed Central) Bd prevalence in field crayfish was **up to 29%.** (PubMed Central) (PNAS)

For your model's γ: No published study has directly measured a transmission coefficient from dead Bd-infected frogs to live susceptible frogs. As an approximation, γ should be substantially lower than α because: (1) carcasses shed zoospores at roughly **1/500th the rate of live infected hosts** (Carter et al. 2024 vs. DiRenzo et al. 2014); (2) cadaver zoospores must still encounter and successfully infect hosts. A reasonable modeling assumption is **$\gamma \approx 0.001 \cdot \alpha$ to $0.01 \cdot \alpha$** (i.e., cadaver transmission 1–10% as efficient as live-host transmission). If $\alpha \approx 10^{-4}$, then $\gamma \approx 10^{-7}$ to 10^{-5}.

δ — Bd on cadavers loses infectivity over 3–12 weeks depending on conditions

The transition from R1 (infectious cadaver) to R2 (non-infectious) depends on environmental conditions, particularly moisture and temperature. Key data:

Johnson & Speare (2003), Emerg. Infect. Dis. 9:922–925, DOI: 10.3201/eid0908.030145 — At 23°C in sterile conditions: "We show that *B. dendrobatidis* will survive in tap water and in deionized water for **3 and 4 weeks**, respectively. In lake water, infectivity was observed for **7 weeks** after introduction." (PubMed Central) This gives:

- Tap water: $\delta \approx 1/21 \approx$ **0.048 day^{-1}**
- Deionized water: $\delta \approx 1/28 \approx$ **0.036 day^{-1}**
- Lake water: $\delta \approx 1/49 \approx$ **0.020 day^{-1}**

Johnson & Speare (2005), Dis. Aquat. Org. 65:181–186 — "Bd is able to survive in the absence of any host species for at least **12 weeks in sterile, moist river sand**, where pH was not limiting." (ResearchGate) This gives $\delta \approx 1/84 \approx$ **0.012 day^{-1}** (longest persistence scenario).

WOAH Manual (2019) — "Zoospores can remain motile for over 24 hours with approximately **50% and 5% remaining motile after 18 hours and 24 hours**, respectively." (WOAH +2) Bd "does not survive desiccation" (WOAH) but "in practice persistence of water in droplets allows survival of the pathogen up to **3 hours after 'drying'**."

Piotrowski et al. (2004), Mycologia 96:9–15, DOI: 10.2307/3761981 — ">95% of zoospores stopped moving within 24 hours." (PubMed) Zoospores swam <2 cm before encysting. (ResearchGate +3) **Bd generation time: 4–10 days** (temperature-dependent). (Springer)

Model-derived values:

- Briggs et al. (2010): Free zoospore mortality **μ = 1 day^{-1}** (1-day lifespan in water) (nih) (PubMed Central)

- Mitchell et al. (2008): Zoospore mortality **μ_Z = 13 yr^{-1}** (~0.036 day^{-1}, ~4-week average lifespan) PubMed
- Wilber et al. (2018, Cambridge Univ. Press book chapter): Environmental zoospore death rates range **2.7×10^{-3} to 4.1×10^{-2} day^{-1}** University of Colorado Bou...

Recommended range for δ: 0.02–0.05 day^{-1} (corresponding to 3–7 week infectious period on cadavers in moist conditions). Use δ ≈ **0.033 day^{-1}** (~30-day half-life) as a central estimate. In dry conditions, δ would be much higher (Bd does not survive desiccation). PubMed Central The Briggs value of μ = 1 day^{-1} applies to free zoospores in water, not to zoosporangia embedded in cadaver tissue, which persist far longer.

η — Antifungal treatment clears Bd in 6–11 days at standard protocols

Brannelly, Richards-Zawacki & Pessier (2012), Dis. Aquat. Org. 101:95–104, DOI: 10.3354/dao02521 — The optimized reduced-dose protocol: "a bath in **0.0025% itraconazole for 5 min day^{-1} for 6 days** reliably cures Bd infection and causes fewer side effects than the longer treatment times and higher concentrations." Inter Research JoVE Species: *Litoria caerulea* and *Incilius nebulifer*. Inter Research Inter Research All animals were Bd-negative 4 weeks post-treatment. PubMed Central **η = 1/6 ≈ 0.167 day^{-1}**.

Nichols & Lamirande (2000/2001), standard protocol — Described in Pessier & Mendelson (2010), Amphibian Ark Disease Manual Chapter 8: **0.01% itraconazole, 5-minute bath daily for 10–11 consecutive days**. Clearance success >90% for post-metamorphic frogs in captive settings. WOAH **η = 1/11 ≈ 0.091 day^{-1}**.

Martel et al. (2011), Med. Mycol. 49:143–149, DOI: 10.3109/13693786.2010.508185 — Voriconazole spray: **1.25 mg/L, once daily for 7 days**. Oxford Academic WOAH **η = 1/7 ≈ 0.143 day^{-1}**.

Hardy et al. (2015), Dis. Aquat. Org. 112:243–250, DOI: 10.3354/dao02813 — Species: *Rana cascadae*, California. Protocol: 0.0025% itraconazole, 5 min/day for 6 days. Results: "Bd prevalence gradually increased in controls but not in treated frogs, with noticeable differences 3 wk after treatment (**27% [4/15] vs. 0% [0/13]**) and strong differences 5 wk after treatment (**67% [8/12] vs. 13% [1/8]**)." Overwinter: **7/30 treated recaptured vs. 0/30 controls**. Inter Research Growth cost: treated frogs weighed **22% less** at 3 weeks and were **9% shorter** at 5 weeks. Inter Research US Forest Service

Cook et al. (2022), Conserv. Sci. Pract. 4:e12762, DOI: 10.1111/csp2.12762 — Species: *Rana cascadae*. Same itraconazole protocol (6 days). Cormack–Jolly–Seber apparent overwinter survival: **itraconazole-treated: 0.48 (95% CI: 0.21–0.86)** vs. **field controls: 0.11 (0.03–0.29)** vs. **experimental controls: 0.02 (0.001–0.06)**. Society for Conservation Bi... "Frogs treated with itraconazole were **more than four times more likely** to survive their first winter."

Society for Conservation Bi... National Wildlife Federation Post-treatment Bd load at 10 months: treated **30.1 ZE** vs. untreated cohort **311.7 ZE** (~10× reduction). Previously infected frogs that survived to adulthood: **5/7 (71%)** treated vs. **0/28** controls. Society for Conservation Bi...

Age-dependent finding (Cook et al. 2022): "Bd prevalence is low in larval Cascades frogs, but newly metamorphosed Cascades frogs experience rapid increases in Bd infection (e.g., from ~10% to ~70% prevalence in 5 weeks)." "Adult Cascades frogs have lower Bd infection intensities than subadult frogs and may experience lower levels of chytridiomycosis-related mortality." ResearchGate

Recommended value for η: 0.091–0.167 day^{-1} (6–11 day treatment duration). Central estimate: **η ≈ 0.167 day^{-1}** (6-day optimized protocol).

ζ — Thermal refugia clear Bd in 5–10 days, with high voluntary shelter use

In your model, ζ represents the rate at which frogs enter the frog sauna (T1). This combines a behavioral component (willingness to use shelter) with the thermal clearance rate once inside. Key data from the landmark frog sauna study:

Waddle et al. (2024), Nature 631:344–349, DOI: 10.1038/s41586-024-07582-y — Species: *Litoria aurea*, Sydney, Australia. Lab thermal gradient experiment: "Six days after being moved into the thermal gradient habitat, the 20 frogs tested had an average infection intensity of **zero**." Popular Science Infected frogs preferred **30°C**; Science preferred body temperature ~**29°C**. The Lighthouse Scientific American In mesocosm: frogs in unshaded (hotter) shelters had **~100-fold lower infection intensity** compared to shaded (cooler) controls. Popular Science Frogs were seen in greenhouses **"four times more often than would be expected by chance."** Science Reinfection study (n=55): pathogen-experienced frogs had **86% survival** vs. naive frogs **22% survival** upon Bd re-exposure Mongabay (**23× odds ratio**). National Geographic

Chatfield & Richards-Zawacki (2011), Dis. Aquat. Org. 94:235–238, DOI: 10.3354/dao02337 — Species: *Rana catesbeiana* and *Acris crepitans*. "**27 of the 28 frogs** that had tested positive were no longer infected with Bd" after **30°C for 10 consecutive days** (**96% clearance**). Conservation Evidence **Clearance rate = 1/10 = 0.10 day^{-1}**.

Woodhams, Alford & Marantelli (2003), Dis. Aquat. Org. 55:65–67, DOI: 10.3354/dao055065 — Species: *Litoria chloris*. "Housing frogs at an environmental temperature of **37°C for less than 16 h** can clear them of the chytrid pathogen." Conservation Evidence PubMed Protocol: 37°C for 8 hours on 2 successive days. AquariaCentral

Stevenson et al. (2013), PLoS ONE 8:e73830, DOI: 10.1371/journal.pone.0073830 — Upper thermal maxima for growth vary by Bd isolate: **26°C (NSW), 27°C (TAS), 28–29°C (QLD)**. "At 29°C, a treatment duration of **8 days** was necessary to prevent recovery of Bd growth, and at 30.5°C a treatment duration of **5 days** was needed."

Piotrowski et al. (2004), Mycologia 96:9–15 — "Exposure of cultures to **30°C for 8 days killed 50%** of the replicates." PubMed Bd grows 4–25°C; optimal 17–25°C. ResearchGate +3

Temperature	Clearance time	Rate (day^{-1})	Source
37°C	<1 day (16h)	~1.0	Woodhams et al. 2003
32°C	5 days	0.20	Retallick & Miera 2007; WOAH
~30°C	10 days	0.10	Chatfield & Richards-Zawacki 2011
~29°C (self-selected)	6 days	0.167	Waddle et al. 2024

For your model's ζ (rate of entering T1): This is a control variable. The Waddle et al. data suggest that when shelters are available, frogs voluntarily use them at **~4× the expected rate.** Science If we assume frogs encounter a sauna once per day on average and use it with 4× preferential probability, and that clearance within the sauna takes ~6–10 days, then the effective ζ (entry rate × clearance probability) is approximately **0.10–0.20 day^{-1}** as a combined parameter. As a pure entry rate (before clearance is factored in), ζ could be treated as a management-adjustable parameter with baseline **0.05–0.20 day^{-1}** depending on shelter density and frog behavior.

ξ — Treatment capture rate depends entirely on management intensity

The rate at which infected frogs enter antifungal treatment (T2) is inherently a management decision variable, not a biological constant. Empirical baselines from field studies:

Harding et al. (2015), Biol. Conserv. — In-situ itraconazole treatment of mountain chicken frogs (*Leptodactylus fallax*) on Montserrat and Dominica. "Multi-state mark–recapture analysis showed increased probability of survival and loss of Bd infection for treated frogs compared to untreated animals." Treatment extended population time to extinction from **49 to 124 weeks** (~60% increase). ScienceDirect Key challenge: "large target population sizes, low capture rates, the potential of reinfection." ScienceDirect

Capture-mark-recapture baselines: Typical field recapture probabilities for frogs are **0.05–0.30 per survey occasion**. With weekly survey frequency:

- Low intensity: ξ ≈ 0.05/7 ≈ **0.007 day^{-1}**
- Medium intensity: ξ ≈ 0.15/7 ≈ **0.021 day^{-1}**
- High intensity: ξ ≈ 0.30/7 ≈ **0.043 day^{-1}**
- Captive colony (daily inspection): ξ ≈ **0.14–1.0 day^{-1}**

Recommended baseline for ξ: 0.01–0.04 day^{-1} for field settings (equivalent to capturing and treating 7–28% of infected frogs per week).

Consolidated parameter table with recommended values

Parameter	Symbol	Recommended range	Central estimate	Units	Primary sources
Transmission rate (live → susceptible)	α	10^{-5} to 10^{-3}	~10^{-4}	frog^{-1} day^{-1}	Briggs 2010; Wilber 2017
Cadaver transmission rate	γ	10^{-7} to 10^{-5}	~10^{-6}	frog^{-1} day^{-1}	Estimated from Carter 2024; Mitchell 2008
Antifungal recovery rate	η	0.091–0.167	0.167	day^{-1}	Brannelly 2012; Hardy 2015
Frog sauna entry/clearance rate	ζ	0.05–0.20	0.10	day^{-1}	Waddle 2024; Chatfield 2011
Disease-induced mortality	β	0.014–0.045	0.033	day^{-1}	Murphy 2011; Murray 2009; DiRenzo 2014
Cadaver Bd decay rate	δ	0.012–0.048	0.033	day^{-1}	Johnson & Speare 2003, 2005
Treatment entry rate	ξ	0.007–0.043	0.021	day^{-1}	CMR field studies

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Critical caveats and modeling considerations

Several issues deserve attention when implementing these parameters. First, **the Briggs et al. (2010) model is not a classical SIR** — it tracks individual Bd loads (sporangia per frog) and a shared zoospore pool. nih PubMed Central Collapsing this to your S-I compartmental structure means α implicitly absorbs zoospore dynamics, making it context-dependent and difficult to transfer across settings. The Briggs model's key insight is that populations can exist in two stable regimes: enzootic (low loads, β ≈ 0) and epizootic (exponentially growing loads, rapid mortality). University of Colorado Bou… PubMed Central Your model should ideally capture this bistability. nih

Second, **age-dependent susceptibility is dramatic**. Cook et al. (2022) and Briggs et al. (2010) both show that metamorphs and subadults face vastly higher mortality than adults. Society for Conservation Bi… In Briggs' model, the lethal sporangia threshold Smax is **10,000 for adults but potentially as low as 1,000 for subadults**. ResearchGate nih Piovia-Scott noted that "adult frogs don't appear to die from it as readily... If we can get them through that vulnerable stage, they apparently do pretty well." The Wildlife Society Hardy et al. (2015) showed Bd prevalence jumps from ~10% to ~70% in untreated metamorphs within just 5 weeks. ResearchGate

Third, **temperature modulates virtually every parameter**. Bd growth ceases above 25–28°C PubMed Central ScienceDirect and the fungus dies above 30°C, ScienceDirect +3 but in vivo temperature effects are host-dependent (Wilber et al. 2016; Raffel et al. 2012). University of Colorado Bou… If your model's environment spans a temperature range, α, β, and the treatment parameters all need temperature-dependent functional forms. The Waddle et al. (2024) frog sauna study demonstrates that enabling behavioral thermoregulation alone can shift the system from epizootic to enzootic dynamics. nature Nature

Fourth, **γ and the R1 compartment represent genuine biology but lack direct empirical estimates for Bd**. The strongest evidence comes from the sister pathogen Bsal, where Carter et al. (2024) demonstrated that carcasses shed ~1,000–1,850 zoospores/day and caused 100% transmission to nearby susceptible hosts. For Bd, Johnson & Speare (2003) showed Bd growing on dead arthropods and algae in water, WOAH and McMahon et al. (2013) demonstrated crayfish as Bd reservoirs. PNAS Bd can persist in moist soil for **up to 12 weeks** ScienceDirect (Johnson & Speare 2005), suggesting the R1 compartment could remain infectious for months under ideal conditions. WOAH However, desiccation rapidly kills Bd, meaning terrestrial cadavers would transition to R2 much faster than aquatic ones.

Finally, the **acquired immunity effect** documented by Waddle et al. (2024) is striking: frogs that cleared Bd via thermal treatment National Geographic were **23× more likely to survive re-exposure**, with 86% vs. 22% survival. This suggests that frogs exiting T1 may warrant a distinct compartment with reduced susceptibility rather than returning directly to S, which would significantly alter long-term model dynamics. The Cook et al. (2022) data similarly show durable protection: treated frogs maintained 10-fold lower Bd loads even 10 months post-treatment.

Conclusion

The parameter landscape for Bd compartmental modeling is rich but heterogeneous. The **most robustly estimated parameter is β** (disease-induced mortality), with convergent evidence from multiple species placing it at **~0.02–0.04 day^{-1}** for acutely susceptible populations. The **least well-constrained parameters are α and γ** (transmission rates), which vary over orders of magnitude depending on host density, water volume, Bd strain, and temperature. Treatment

parameters (η, ξ) are well-defined by established veterinary protocols but represent management choices rather than fixed biology. The frog sauna parameter (ζ) benefits from the Waddle et al. (2024) study showing voluntary shelter use and 6-day clearance, but will vary with shelter design, density, and local climate. For initial model exploration, the central estimates in the consolidated table above provide a defensible starting point, with sensitivity analysis across the full ranges being essential given the multi-order-of-magnitude uncertainty in transmission parameters.